

A COUNTEREXAMPLE TO A SELF-ADHESIVITY CONJECTURE OF BOEGE, BOLT, AND STUDENÝ

ABSTRACT. Boege, Bolt, and Studený asked whether

$$\mathbf{sg}^\diamond(N) \cap \mathbf{st}(N) = \mathbf{st}^\diamond(N)$$

holds for every finite variable set N . We give an explicit structural semi-graphoid on five variables that is self-adhesive relative to the semi-graphoid frame but not relative to the structural frame. This answers their Open Problem 3 in the negative and, by the four-variable computation of Boege, Bolt, and Studený, five variables are minimal. The proof consists of an integer supermodular witness, exact semi-graphoid closure computations, and a short linear certificate for the failure of structural self-adhesivity.

1. STATEMENT AND NOTATION

Let $\mathbf{sg}(N)$ and $\mathbf{st}(N)$ denote, respectively, the semi-graphoids and structural semi-graphoids on a finite variable set N , and let $\mathbf{f}^\diamond(N)$ denote the models self-adhesive relative to a frame $\mathbf{f}$; we use the definitions of [1]. Since $\mathbf{st} \subseteq \mathbf{sg}$, anti-extensivity and isotonicity of the self-adhesion operator [1, Lemma 5.2] give

$$\mathbf{st}^\diamond(N) \subseteq \mathbf{sg}^\diamond(N) \cap \mathbf{st}(N).$$

Boege, Bolt, and Studený verified equality computationally for $|N| = 4$ and asked whether it holds in general. Their question, Open Problem 3 of [1] and the question closing their Section 7.4, reads: “We would like to know whether the relation $\mathbf{sg}^\diamond(N) \cap \mathbf{st}(N) = \mathbf{st}^\diamond(N)$ holds in general.”

Theorem 1. *For $N = \{0, 1, 2, 3, 4\}$,*

$$\mathbf{st}^\diamond(N) \subsetneq \mathbf{sg}^\diamond(N) \cap \mathbf{st}(N).$$

Equivalently, the conjectural equality in [1, Open Problem 3, the question closing Section 7.4] fails on five variables, and this is the smallest possible cardinality.

An elementary conditional-independence statement is written $ij \mid K$, where $i, j \in N$ are distinct and $K \subseteq N \setminus \{i, j\}$. For a set function $f: 2^N \rightarrow \mathbb{R}$, write

$$\Delta_f(ij \mid K) := f(ijK) + f(K) - f(iK) - f(jK).$$

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The function f is supermodular when all elementary differences are nonnegative, and it induces the structural semi-graphoid

$$\mathcal{M}_f := \{ij \mid K : \Delta_f(ij \mid K) = 0\}.$$

For pairwise disjoint A, B, C , let $[A, B \mid C]$ denote the set of all $ij \mid K$ with $i \in A, j \in B$, and $C \subseteq K \subseteq ABC \setminus \{i, j\}$.

2. THE COUNTEREXAMPLE

Let

$$\begin{aligned} \mathcal{M} = \{ & 01 \mid \emptyset, 01 \mid 4, 01 \mid 23, 01 \mid 24, 01 \mid 234, 02 \mid 13, 03 \mid 4, 03 \mid 12, 04 \mid 12, \\ & 12 \mid \emptyset, 12 \mid 03, 13 \mid \emptyset, 13 \mid 02, 14 \mid \emptyset, 14 \mid 0, 23 \mid 01, \\ & 24 \mid \emptyset, 24 \mid 0, 24 \mid 3, 34 \mid 0\}. \end{aligned}$$

Juxtaposition denotes union. Define the standardized integer-valued set function $f: 2^N \rightarrow \mathbb{Z}$ by

$f(S)$	S
0	$ S \leq 1, 01, 12, 13, 14, 24$
1	$02, 03, 23, 34$
2	$04, 014, 234$
3	$012, 013, 123, 024, 124, 034, 134$
4	023
6	$0123, 0124$
7	$0134, 0234, 1234$
12	01234

Lemma 2. *The function f is supermodular and $\mathcal{M}_f = \mathcal{M}$. Hence $\mathcal{M} \in \mathbf{st}(N)$.*

Proof. There are $\binom{5}{2}2^3 = 80$ elementary differences. Exact evaluation gives

$$(\#\{s : \Delta_f(s) = r\})_{r=0}^4 = (20, 20, 25, 14, 1),$$

and the twenty zero differences are precisely the statements listed in $\mathcal{M}$. Thus every elementary difference is nonnegative and $\mathcal{M}_f = \mathcal{M}$. $\square$

Remark 3. *The twenty active elementary equalities have rank 19 in the 26-dimensional space of standardized set functions. Thus f lies in a seven-dimensional face of the supermodular cone. The coatoms of $(\mathbf{st}(N), \subseteq)$ correspond to the extreme rays of the cone of standardized supermodular functions, and a seven-dimensional face is not an extreme ray. Hence this model is not a coatom, and it lies outside the 1319 permutational types of five-variable coatoms tested in [1, Section 7.6].*

Lemma 4. *The model $\mathcal{M}$ belongs to $\mathbf{sg}^\diamond(N)$.*

Proof. Fix $L \subseteq N$ and a canonical copy $b_L: N \rightarrow N_L$ fixing L . Put

$$\mathcal{A}_L := \mathcal{M} \cup b_L(\mathcal{M}) \cup [N \setminus N_L, N_L \setminus N \mid L].$$

By Remark 1 after Definition 4.3 and Lemma 4.5 of [1], one canonical copy suffices and

$$\mathfrak{sg}^\diamond(\mathcal{M} \mid L) = \mathfrak{sg}(\mathcal{A}_L) \downarrow N,$$

where $\mathfrak{sg}(\mathcal{A}_L)$ is the closure under the elementary semi-graphoid rule

$$\{ij \mid K, i\ell \mid jK\} \iff \{i\ell \mid K, ij \mid \ell K\}.$$

Corollary 4.12 of [1] covers $|L| \leq 1$, and Lemmas 4.15–4.16 there cover the co-singletons $|L| = 4$; the semi-graphoid frame satisfies the hypotheses of these results, and of Lemma 4.5, by [1, Lemmas 3.1, 4.10, and 4.15]. With the trivial case $L = N$, only $|L| \in \{2, 3\}$ requires computation. Exact saturation gives the following sizes; in every row the N -marginal of the closure is exactly the twenty-statement model $\mathcal{M}$.

L	$ \mathcal{A}_L $	$ \mathfrak{sg}(\mathcal{A}_L) $	L	$ \mathcal{A}_L $	$ \mathfrak{sg}(\mathcal{A}_L) $
01	183	430	012	54	172
02	184	494	013	54	154
03	184	437	014	52	52
04	184	577	023	56	169
12	183	419	024	54	81
13	183	374	034	54	67
14	183	346	123	54	165
23	184	405	124	53	108
24	183	248	134	54	67
34	184	325	234	54	60

Therefore $\mathfrak{sg}^\diamond(\mathcal{M} \mid L) = \mathcal{M}$ for every $L \subseteq N$, and $\mathcal{M} \in \mathfrak{sg}^\diamond(N)$. $\square$

Lemma 5. *The model $\mathcal{M}$ does not belong to $\mathfrak{st}^\diamond(N)$.*

Proof. Take $L = 012$, let $N' = \{0, 1, 2, 5, 6\}$, and define the copy $b: N \rightarrow N'$ by

$$b(0) = 0, \quad b(1) = 1, \quad b(2) = 2, \quad b(3) = 5, \quad b(4) = 6.$$

Suppose that a structural adhesion exists. It is induced by a supermodular function $g: 2^{\{0, \dots, 6\}} \rightarrow \mathbb{R}$. Write $\delta_{ij|K} := \Delta_g(ij \mid K)$. Exact equality of the two marginals and the adhesion condition imply

$$\delta_s = 0 \quad \text{for every } s \in \mathcal{M} \cup b(\mathcal{M}) \cup [34, 56 \mid 012].$$

A coefficientwise expansion of elementary differences gives the identity

$$\begin{aligned}
 (1) \quad & \delta_{04|3} + \delta_{01|235} + \delta_{01|245} + \delta_{05|234} + \delta_{15|0234} \\
 & + \delta_{34|25} + \delta_{35|02} + \delta_{35|12} + \delta_{45|02} + \delta_{45|12} \\
 = & \underbrace{\delta_{01|23} + \delta_{04|12} - \delta_{24|0} + \delta_{24|3} + \delta_{34|0}}_{\text{statements in } \mathcal{M}} \\
 & + \underbrace{\delta_{01|25} + \delta_{05|12} + \delta_{15|02}}_{\text{statements in } b(\mathcal{M})} \\
 & + \underbrace{2\delta_{35|012} + \delta_{45|012} + \delta_{45|0123}}_{\text{statements in } [34,56|012]}.
 \end{aligned}$$

The right-hand side is zero. Every term on the left-hand side is nonnegative by supermodularity, so all of them vanish. In particular, $\delta_{04|3} = 0$, and the marginal on N contains $04 \mid 3$. This contradicts the required marginal $\mathcal{M}$, because $04 \mid 3 \notin \mathcal{M}$; indeed,

$$\Delta_f(04 \mid 3) = 1.$$

□

Proof of Theorem 1. Lemmas 2, 4, and 5 give

$$\mathcal{M} \in \mathfrak{sg}^\diamond(N) \cap \mathfrak{st}(N) \setminus \mathfrak{st}^\diamond(N).$$

For $|N| \leq 3$, the two frames coincide and are fixed by self-adhesion [1, Corollary 4.17]; for $|N| = 4$, the asserted equality was established in [1, Section 7.4]. The case $|N| = 4$ rests on the SAT and LP computations reported there rather than on a proof, and we inherit it as stated. Hence five variables are minimal. □

EXACT VERIFICATION

All computations above are determined by the data printed here and are reproducible in exact integer and rational arithmetic. From the table defining f one evaluates the 80 elementary differences $\Delta_f(ij \mid K)$ and compares their multiset with the histogram in Lemma 2 and their zero set with $\mathcal{M}$. The rank assertion of Remark 3 is a row reduction of the twenty corresponding equality functionals in the 26 coordinates indexed by the subsets of N of size at least two. For each of the twenty sets L with $|L| \in \{2, 3\}$, the set $\mathcal{A}_L$ is read off from $\mathcal{M}$ and saturated under the elementary semi-graphoid rule on $N \cup N_L$, which returns the sizes tabulated in Lemma 4 and the N -marginal $\mathcal{M}$. Finally, identity (1) is an identity between integer combinations of the 128 values of g and is checked by expanding both sides coefficient by coefficient. No floating-point computation is involved.

REFERENCES

- [1] T. Boege, J. H. Bolt, and M. Studený, *Self-adhesivity in lattices of abstract conditional independence models*, Discrete Appl. Math. **361** (2025), 196–225. doi:10.1016/j.dam.2024.10.006. Numbering of results and open problems above follows the authors' e-print arXiv:2402.14053v3.